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Weekly update 4 of 5 · week of Nov 30

Prompt v3 and a held-out check

THIS UPDATE IN THREE NUMBERS

25.0%

coverage (v1 19.5%, v2 15.1%)

0.86

bias (v1 0.75): half of the underestimate removed

12.5%

run-to-run CV (v1 19.0%)

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What prompt v3 adds (on top of the v2 scope and method)

1 Sourced intensity table

A1–A3 per m² of component, compiled by a separate research subagent from external sources (CLF 2025 baselines, PCI, Kawneer, RDH); **excludes the ground-truth dataset**. Placed at the step where numbers are chosen.

2 XPS modelling convention

LCA tools of the period often used **HFC-blown XPS** for roofs.

3 Sanity check

Roughly **50 kgCO_2e/m² of envelope** as a cross-check of the total.

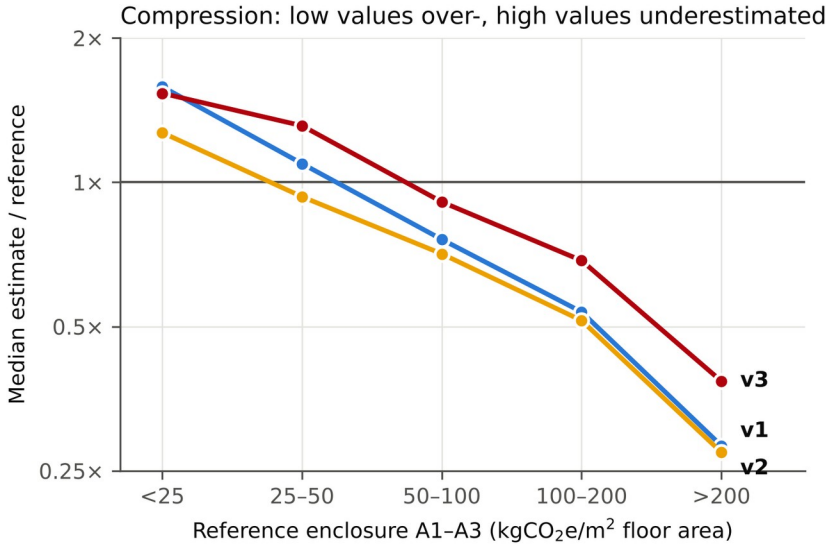
4 Renovation rule

Rewritten so that it **no longer subtracts** by default.

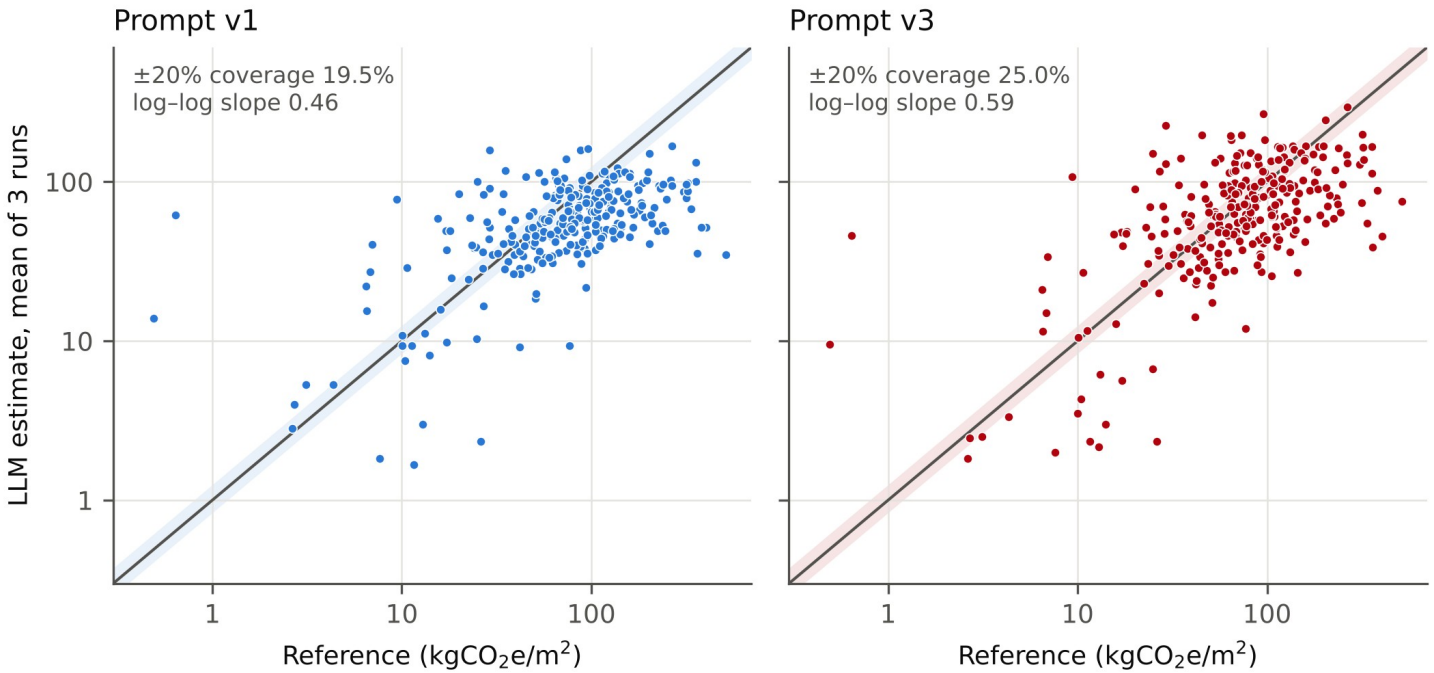
Results: all three prompts (292 buildings, 3 runs each)

Prompt	Coverage	RMSE	Med. APE	Med. |log|	Bias	CV
v1	19.5%	79.7	37.9%	0.456	0.75	19.0%
v2	15.1%	81.0	44.3%	0.563	0.67	13.8%
v3	**25.0%**	**76.0**	**40.0%**	**0.426**	**0.86**	**12.5%**

- v3 vs v2: **significant** (p = 0.002). v3 vs v1: consistent but **borderline** (McNemar p = 0.085, Wilcoxon p = 0.053).
- HFC-XPS is now considered in **96%** of answers for projects with XPS.
- Unbiased now: the non-XPS part (0.99), buildings with a given envelope area (0.99), warm zones (0.98).

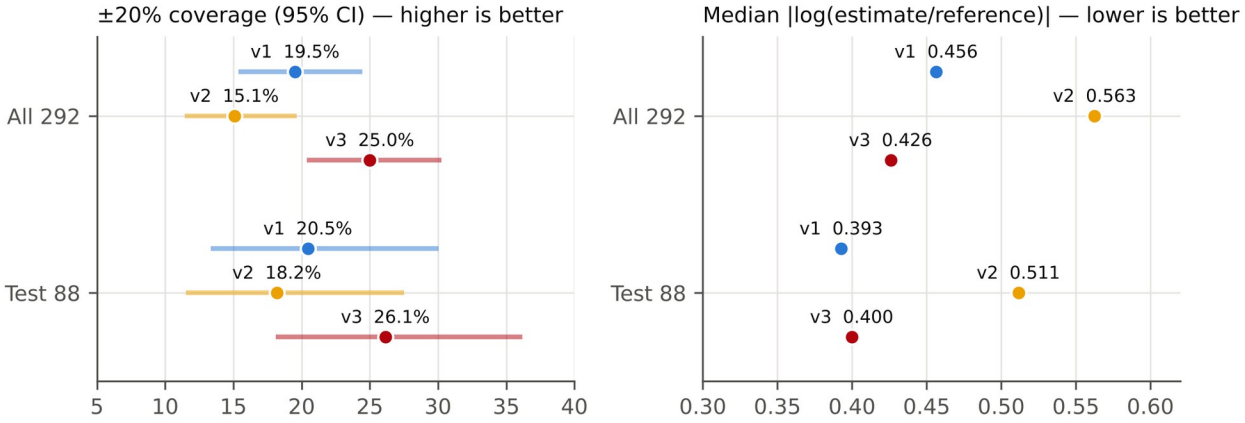


What is still wrong



- **Compression at the top:** 0.38 above 200 $kgCO_2e/m^2$.
- **Light enclosures** are overestimated more (1.31 in the 25–50 bin).
- **Renovations** remain hard: the extent of envelope work is not in the features.

Held-out check: 88 test buildings



- **No external test set fits:** the CLF benchmark probably shares source models; CarbEnMats has 172 European buildings, a different element definition and no geometry.
- **Split:** 88 test / 204 dev, stratified on project type × climate (seed 226). It was **assigned after the runs**, so we re-ran every diagnostic on dev only, and all findings replicate (slope 0.44, XPS 14.7%, area effect 0.79 vs 0.65, polyiso 89%).
- **Test:** v3 26.1% vs v1 20.5% coverage, RMSE 69.2 vs 74.1, bias 0.86 vs 0.75. Median |log| unchanged (0.400 vs 0.393); **not significant** at this size (p = 0.41).

Next update

Coming next

- **Feature study:** which inputs does the model actually need?
- **LEED v5:** what the accuracy means for MRp2 and MRc2.
- Final deliverables: paper, final prompt, poster.

Input we'd like

- Is the **post-hoc split with dev-only replication** acceptable, or should we collect additional projects?